Linear Modelling

Questions 1–10

Question 1. For each of the following situations, write an equation to model the information:

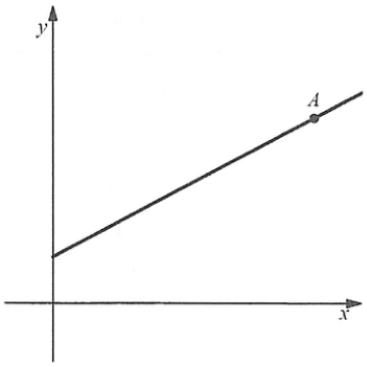
a) When making pastry, the weight of fats (y) needed is half the weight of the flour (x)
b) A business adds a tax of 20% to all items it sells (x) before calculating its sales price (y)
c) For each £1 converted into US dollars (x) you receive \$1.29 (y)

Question 2. Three phone companies charge mobile phone tariffs usually known as 'pay-as-you-go' which charge £C the user per t minutes and have no monthly cost.

Phone Company	Standard Charge (per minute)
H_2O	25p
UU	30p
PiffPaff	24p

a) On the same axes, sketch a graph representing the cost over time for making calls using each of the companies. Label the point at which £3 has been spent.
b) In each case write down an equation for C in terms of t.

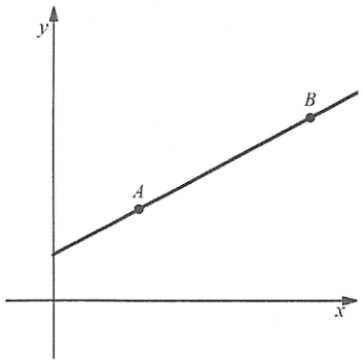
Question 3. The graph below shows a straight-line graph which goes through the point A.



Find the equation of the line given

a) Point A is $(1, 9)$ and the y-intercept is 8
b) Point A is $(5, 4)$ and the y-intercept is 2
c) Point A is $(3, 7)$ and the y-intercept is 5

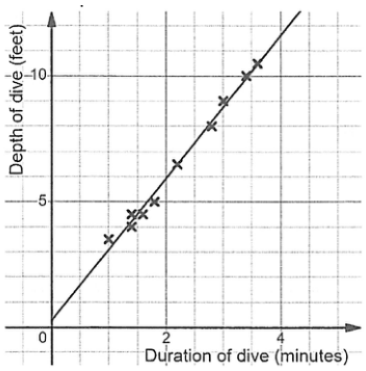
Question 4. The graph below shows a straight-line graph going between two points A and B.



Find the equation of the line in the form $y = mx + c$ given points A and B are

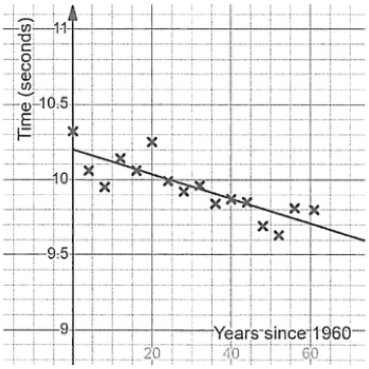
a) $(2, 10)$ and $(5, 19)$
b) $(1, 2.5)$ and $(7, 5.5)$
c) $(3, 7)$ and $(15, 12)$

Question 5. Data has been collected on the dive of penguins. The depth of their dives d in feet and duration of their dive duration in time t minutes is plotted below.



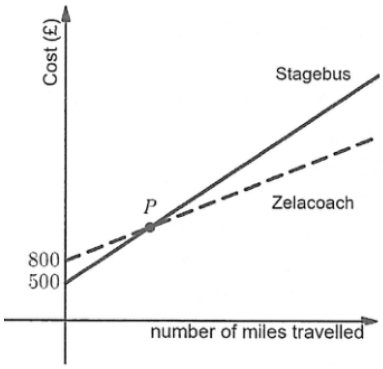
a) Use the graph to write an equation of d in term of t
b) Use your equation to predict how deep a penguin will dive in a 2.5 minute dive.
c) What does the gradient of the graph represent? State the units.
d) What does the vertical intercept represent in context?
e) Comment on the validity of the vertical intercept.

Question 6. The graph below shows the times of the gold medallists, T seconds, in the men's 100 metre event in the Olympics against the years since 1960, x years. The line is given by the equation $T = -0.0082x + 10.2$.



a) According to the model, what time will someone in the 2028 Olympics achieve?
b) Explain what the 10.2 and the -0.0082 represents in context.
c) Explain why a linear model is not valid in the long term.

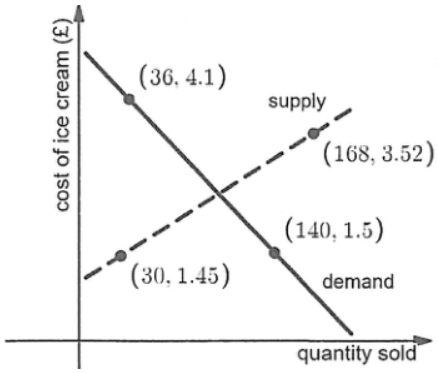
Question 7. The graph below shows the cost £C of hiring a coach for the number of miles travelled m from two different companies, Stagebus and Zelacoach.



For a distance of 220 miles, Stagebus quotes a cost of £2000 whilst Zelacoach charges £1700.

a) Explain what the values on the vertical axis represent in context for each company.
b) Find an equation that models the costs of hiring coaches from the two companies.
c) Identify the coordinates of the point P labelled on the graph and state the relevance of this point in context.

Question 8. The graph below shows the link between the price of an ice cream £P and the number of them sold x in the shop.



State the point at which the two lines intersect and interpret what that point means in context.

Question 9. Two companies charge different costs for electrical work at their business premises. **Electrosolutions:** A call out charge plus £30 an hour after that. **Ohms Electrical:** Starting at £90 covering the first 2 hours and then following a linear relationship with an hourly rate. Rebecca uses both companies for her business and finds that they both charged £170 for 4 hours work.

a) Find the call out charge of Electrosolutions.
b) Find the hourly rate of Ohms Electrical.
c) Ohms Electrical want to change to a call out charge with their current hourly rate. What would be their initial call out charge?

Question 10. A bakery makes muffins. They sell the muffins at £2.00 each. There is an upfront cost in making the muffins, plus a fixed cost per muffin. When 500 muffins are made and sold, a profit of £300 is made, but when 100 muffins are made and sold, there is a loss of £50.

a) Show that the cost £C in terms of the number of muffins x, may be modelled as $C = 1.125x + 137.5$.
b) Find the minimum number of muffins that need to be made and sold for the bakery to not make a loss.